Predicting Survival of Patients with Heart Failure with Machine Learning Methods

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1 Abstract

This reproduction study investigates the use of random forest classification models to predict survival among patients with heart failure based on clinical indicators. This study reproduces the work published by Jia (2024). Two models were developed: one using only serum creatinine, ejection fraction, and follow-up time as predictors, and a second including all quantitative variables. The results showed that the model with the narrower set of predictors performed better, achieving an accuracy of 80%, compared to 75% for the model using all quantitative predictors. Feature importance analysis revealed that follow-up time was the most influential predictor, followed by serum creatinine and ejection fraction. These findings align with Jia (2024), where serum creatinine and ejection fraction were identified as key predictors of heart failure outcomes. However, the higher accuracy of the simpler model suggests that including additional predictors may introduce noise rather than improving predictive performance. This study highlights the potential of machine learning approaches, particularly random forests, in enhancing survival prediction and guiding clinical decision-making in heart failure management.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import StratifiedKFold, KFold

df = pd.read_csv('heart_failure_clinical_records_dataset.csv')
```

2 Introduction

Approximately 6.7 million people in the United States, aged 20 years or older are diagnosed with heart failure, a condition that occurs when the heart is incapable of pumping adequate blood and oxygen to organs in the body (CDC, 2024). In 2022, heart failure was mentioned on nearly 500,000 death certificates and was responsible for 13.9% of causes of death, nationwide (CDC, 2024). This widespread effect of heart failure emphasizes the need for research in new technologies and/or interventions to minimize the impact of the disease.

With the rise of machine learning, making accurate predictions rather than relying on explantory frameworks has received greater attention in recent years (Yarkoni & Westfall, 2017). Scientists argue that good explanation begins with good prediction (Yarkoni & Westfall, 2017). One practical application of this is in the work of Chicco & Jurman, who use machine learning methods to predict the survival of patients with heart failure based on ejection fraction and serum creatinine with high accuracy (Chicco & Jurman, 2020). This work was later reproduced and published by Jia (Jia, 2024). The current study seeks to reproduce parts of the results of Jia's reproduction, specifically the random forest model. This project ensures the reliability and generalizability of the random forest model in predicting heart failure survival using serum creatinine and ejection fraction.

2.1 Dataset

The dataset for this study was obtained from the UC Irvine Machine Learning Repository, which is a widely used source for benchmarking machine learning models. The dataset includes clinical records of heart failure patients, with features such as age, serum creatinine, ejection fraction, serum sodium, platelet count, and creatine phosphokinase levels. These variables were selected based on their established relevance in heart failure prognosis (Chicco & Jurman, 2020). Data preprocessing involved handling missing values, normalizing continuous variables, and splitting the dataset into training and testing subsets to evaluate model performance. By leveraging this dataset, the current study aims to assess the reproducibility of Jia's findings and validate the predictive power of random forest classification in this context.

3 Methods

3.1 Participants

299 patients in the Heart Failure Clinical Reocrds dataset were categorized into two groups: deceased patients (patients who did not survive) and survived patients (patients who were still alive at the end of the observation period). The dataset was summarized by computing mean, median, and standard deviation for the full sample, as well as for the deceased and survived patient groups separately. The summary statistics (see Table X) provide insights into the distribution of key clinical variables across survival outcomes.

3.2 Descriptive Statistics

The average age of the full sample was 60.83 years, with deceased patients being older on average (65.22 years) compared to those who survived (58.76 years). The mean serum creatinine level was 1.39 mg/dL for the full sample, but significantly higher in deceased patients (1.84 mg/dL) than in survivors (1.18 mg/dL), indicating a potential link between renal function and mortality risk. The mean ejection fraction for the full sample was 38.08%, but lower among deceased patients (33.47%) compared to survivors (40.27%), suggesting that reduced ejection fraction is associated with increased mortality risk. Creatinine phosphokinase, an enzyme indicative of muscle damage, showed considerable variability with higher average values in deceased patients (670.20 U/L) compared to survivors (540.05 U/L). The mean serum sodium level was slightly lower in deceased patients (135.38 mmol/L) than in survivors (137.22 mmol/L). The recorded follow-up duration was notably shorter for deceased patients (70.88 days) compared to survivors (158.34 days), reflecting the difference in survival time between the two groups.

3.3 Ejection Fraction

Ejection fraction in the context of the heart is a measure of the ratio of blood pumped out of the heart's left ventricle at each contraction (Foley et al., 2012). Patients with reduced ejection fraction form a well-defined subset of cases, highlighting its central role in heart failure classification and prognosis (Iwano & Little, 2013). Incorporating ejection fraction into machine learning models enhances predictive accuracy by allowing algorithms to capture these nuanced distinctions in patient outcomes.

Ejection fraction is calculated by first measuring end-diastolic-volume (EDV), which is the amount of blood collected in the left ventricle just prior to contraction (Foley et al, 2012). Then another measurement of blood in the left ventricle is collected immediately after contraction, known as end-systolic-volume (ESV) (Foley et al., 2012). From this, a stroke volume (SV) is calculated by subtracting ESV from EDV, and then dividing by EDV (Foley at al., 2012). This results in a decimal, which can be multiplied by 100 to obtain an ejection fraction percentage.

$$SV = EDV - ESV$$

$$EF(\%) = \frac{SV}{EDV} \times 100$$

Where:

- (EF) = **ejection fraction**: Proportion of blood pumped out of the left ventricle of the heart (%).
- (EDV) = **end-diastolic volume**: Amount of blood within the left ventricle prior to contraction (mL).
- (ESV) = **end-systolic-volume**: Amount of blood withiin the left ventricle immediately after contraction (mL).
- (SV) = **stroke volume**: Amount of blood pumped out of the left ventricle (mL).

3.4 Serum Creatinine

Serum creatinine refers to the amount of creatinine in a patient's blood, measured in milligrams per deciliter. Creatinine is a waste product that accumulates from muscle use and breakdown, and is then deposited into the bloodstream (Johns Hopkins Health Plans, 2022). From there, creatinine travels to the kidneys via the bloodstream, where healthy kidneys would filter and dispose of the creatinine build-up, circulating waste-free blood through to the rest of the body (Johns Hopkins Hleath Plans, 2022).

Patients with heart failure often have elevated serum creatinine levels (Shlipak et al., 2003). Research has shown that heart failure patients with elevated serum creatinine levels tend to have poorer disease outcomes (Shlipak et al., 2022). Including serum creatinine in a machine learning model allows the model to capture the impact of kidney dysfunction on survival outcomes, improving predictive accuracy and clinical relevance.

3.5 Follow-Up Time

Follow-up time was included as a predictor to account for the duration between a patient's initial assessment and the recorded outcome. This variable provides important temporal context, capturing how survival likelihood may change over time. Data on follow-up time were extracted from patient records and measured in months. Given its potential influence on survival outcomes, follow-up time was considered alongside other clinical indicators in the model to assess its contribution to predictive accuracy.

```
df.columns = df.columns.str.lower()

quant_preds = ['age', 'creatinine_phosphokinase', 'ejection_fraction',
               'platelets', 'serum_creatinine', 'serum_sodium', 'time']

full_stats = df[quant_preds].agg(['mean', 'median', 'std'])
death_stats = df[df['death_event'] == 1][quant_preds].agg(['mean', 'median', 'std'])
survival_stats = df[df['death_event'] == 0][quant_preds].agg(['mean', 'median', 'std'])

combined_stats = pd.concat([full_stats, death_stats, survival_stats],
                           keys=['Full Sample', 'Deceased Patients',
                                 'Survived Patients']).transpose()
```

	('Full Sample', 'mean')	('Full Sample', 'median')	('Full Sample', 'std')	('Deceased Patients', 'mean')	('Deceased Patients', 'median')	('Deceased Patients', 'std')	('Survived Patients', 'mean')	('Survived Patients', 'median')	('Survived Patients', 'std')
age	60.8339	60	11.8948	65.2153	65	13.2146	58.7619	60	10.6379
creatinine_phosphokinase	581.839	250	970.288	670.198	259	1316.58	540.054	245	753.8
ejection_fraction	38.0836	38	11.8348	33.4688	30	12.5253	40.266	38	10.86
platelets	263358	262000	97804.2	256381	258500	98525.7	266657	263000	97531.2
serum_creatinine	1.39388	1.1	1.03451	1.83583	1.3	1.46856	1.18488	1	0.654083
serum_sodium	136.625	137	4.41248	135.375	135.5	5.00158	137.217	137	3.98292
time	130.261	115	77.6142	70.8854	44.5	62.3783	158.34	172	67.7429

3.6 Data Analysis

To better understand the patterns above and make accurate predictions, this study reproduced a random forest classification model. This method was chosen due to its ability to handle nonlinear relationships, account for interactions among variables, and provide insights into feature importance. The following section details the setup and evaluation of the random forest models.

Random forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their outputs to improve predictive accuracy and reduce overfitting. Each tree is trained on a random subset of the data, and the final classification is determined by majority vote among the trees. Random forests can be an excellent way to explore nonlinear patterns in complex datasets. In this study, a random forest classifier was implemented to predict patient survival based on key clinical indicators, including serum creatinine and ejection fraction. Two models were run in this reproduction. First, a random forest classifier was set up using only serum creatinine and ejection fraction as predictors of a death event. A second model was run using all quantitative variables as predictors in the model.

3.7 Random Forest w/ all Quantiative Predictors

The first random forest model run in this study was set up with all quantitative predictors. Quantitative predictors were isolated and transformed using the StandardScaler from scikit-learn. The features were

standardized using the StandardScaler from scikit-learn, ensuring that they were on the same scale. Scaling features improves the performance of many machine learning algorithms, producing more stable results. The data were then split into training and testing sets, with 30% of the data allocated to the test set using the train_test_split function. This split helps evaluate the model's generalizability to unseen data. The random forest classifier was instantiated and trained on the training data with a random state of 42 for consistency. Finaly predictions were made and compared to the test set to assess the model's performance in a new dataset.

```
the_scaler = StandardScaler().set_output(transform="pandas")
X = the_scaler.fit_transform(df[quant_preds])
y = df['death_event']
```

```
X_train, X_test, y_train, y_test = train_test_split(
     X, y, test_size=0.33, random_state=42)
```

```
the_rf2 = RandomForestClassifier()
the_rf2.fit(X_train, y_train)
preds = the_rf2.predict(X_test)
```

```
(y_test == preds).mean()
```

```
0.7373737373737373
```

3.8 Feature Importance

Feature importances were extracted from the random forest model containing all quantitative predictors using the .feature_importances function. This give a measure of the weight each predictor in the model's predictions. Feature importances can also be respresented visually to demonstrate the each predictor's impact on the model.

```
the_rf2.feature_importances_
```

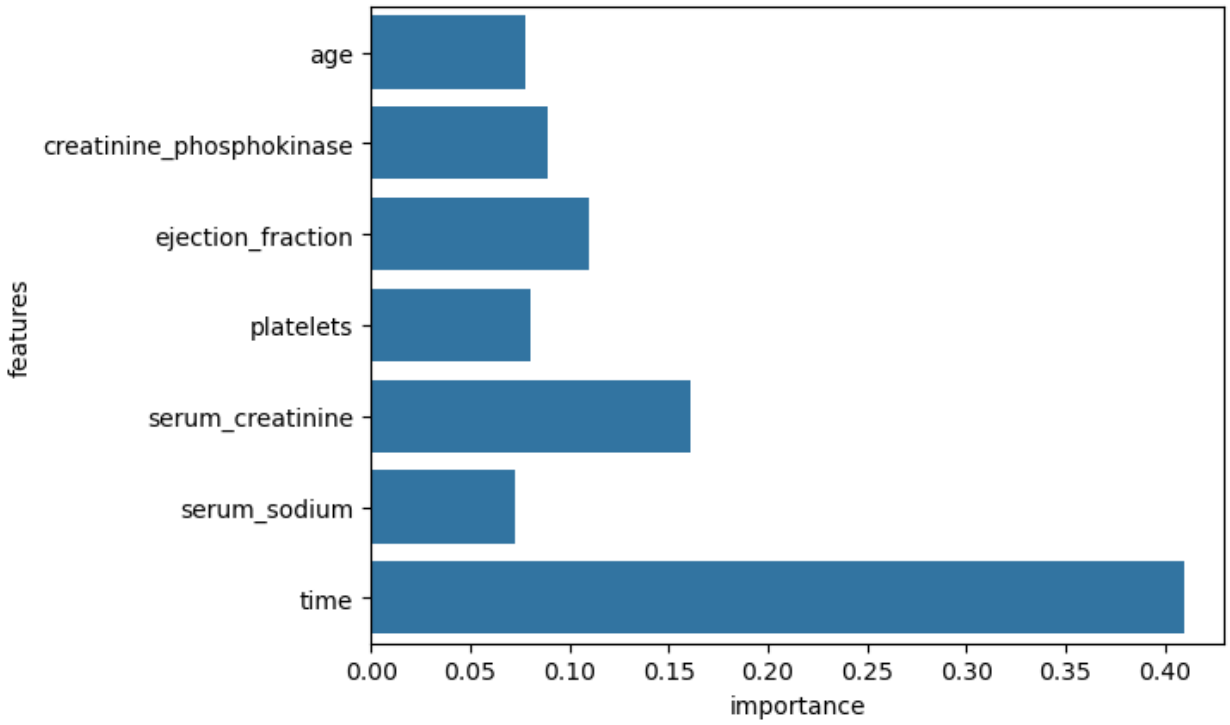
```
array([0.07752675, 0.08894009, 0.11025899, 0.0805131 , 0.16083354,
       0.07263075, 0.40929678])
```

```
viz_fi = pd.DataFrame({'features': X.columns, 'importance': the_rf2.feature_importances_})
viz_fi.style.hide()
```

Table 2

features	importance
age	0.077527
creatinine_phosphokinase	0.088940
ejection_fraction	0.110259
platelets	0.080513
serum_creatinine	0.160834
serum_sodium	0.072631
time	0.409297

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.barplot(viz_fi, x = 'importance', y = 'features')
```



3.9 Random Forest Classification: Ejection Fraction & Serum Creatinine

In this second analysis, primary features from the first random forest model were first isolated based on the feature importance results from above: serum creatinine, ejection fraction, and follow-up time. All three feautres were isolated and transformed using the StandardScaler from scikit-learn. The transformed data was then split into training and testing sets, with the same 70/30 split used in the previous model. A random rorest classifier was instantiated and trained on the training data and a random state of 42 was set. After fitting the model, predictions were made on the test set using the predict method. These predictions were then compared against the actual test set values to assess the model's performance.

```
primary_features = ['serum_creatinine', 'ejection_fraction', 'time']

the_scaler = StandardScaler().set_output(transform="pandas")
X = the_scaler.fit_transform(df[primary_features])
y = df['death_event']
```

```
X_train, X_test, y_train, y_test = train_test_split(
     X, y, test_size=0.33, random_state=42)
```

```
the_rf1 = RandomForestClassifier()
the_rf1.fit(X_train, y_train)
preds = the_rf1.predict(X_test)
```

```
(y_test == preds).mean()
```

```
0.797979797979798
```

4 Results

In this study, two random forest models were trained to predict patient survival based on clinical indicators. The first model, which included all quantitative variables, showed an accuracy of 74%, which is similar to

the results reported by Jia (2024), which were accuracy rates of 69% and 78% in their stratified k-fold and k-fold models, respectively. The second model used only ejection fraction and serum creatinine as predictors, achieving an accuracy of 80%%. This result is higher than those report by Jia (2024), which were 65% and and 78% for their stratified k-fold model and k-fold model, respectively. These findings suggest that a simplified model with just ejection fraction and serum creatinine may offer improved predictive performance for patient survival compared to models using a broader set of clinical indicators.

The feature importance analysis revealed that the most influential predictors in the model were follow-up time (0.41), serum creatinine (0.16), and ejection fraction (0.11). Serum creatinine and ejection fraction were notably higher in importance compared to other features, aligning with their established clinical relevance in predicting heart failure outcomes. Follow-up time was the most important predictor, with an importance score of 0.41, reflecting its central role in tracking patient condition over time. Follow-up time is a feature that was not included in Jia's feature importance, interestingly. Other clinical measures, such as age, creatinine phosphokinase, and serum sodium, contributed to the model but with lower importance scores.

Comparing these results to Jia's (2024) study, which also utilized random forest models to predict heart failure survival, findings od this study are mostly consistent with Jia's, where serum creatinine and ejection fraction were found to be significant predictors of survival. Jia's model reported similar trends in feature importance, particularly highlighting serum creatinine and ejection fraction as key contributors. However, Jia (2024) included a model with thre three mos important features, but did not contain one of those features (follow-up time) in their feature importance. The higher accuracy of the model with fewer features highlights the potential benefits of a more streamlined feature set. Additionally, this finding might suggest simplifying the data collection process, indicating that collecting fewer, more important features has the potential to produce more accurate prediction regarding survival rates in patients with hear failure.

These results highlight the effectiveness of machine learning, particularly random forests, in improving survival predictions for heart failure patients, with both models outperforming the original study's stratified k-fold results.

5 Discussion

This study demonstrates the effectiveness of random forest models in predicting patient survival in the context of heart failure. By comparing two models—one incorpoating all quantiative predictors as features, and the other using only serum creatinine, ejection fraction, and follow-up time—we observed a significant improvement in accuracy from 74% to 80% with the inclusion of fewer, more important predictors. The feature importance analysis revealed that serum creatinine and ejection fraction were the most influential predictors, consistent with Jia's (2024) study. These findings suggest that a more streamlined model focusing on the most relevant clinical indicators can improve predictive accuracy while reducing complexity. Future work should explore whether additional simplifications or alternative feature selection methods can further enhance the performance of survival prediction models in heart failure cases.

6 References

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7 Appendix A

Below is a data dictionary containing all variables contained in Heart Failure Clinical Records Dataset, their descriptions, units or measurements, and ranges.

```
data_dict = pd.DataFrame({
    "Feature": [
        "Age", "Anaemia", "High blood pressure", "Creatinine phosphokinase (CPK)",
        "Diabetes", "Ejection fraction", "Sex", "Platelets", "Serum creatinine",
        "Serum sodium", "Smoking", "Time", "Death event"
    ],
    "Explanation": [
        "Age of the patient",
        "Decrease of red blood cells or hemoglobin",
        "If a patient has hypertension",
        "Level of the CPK enzyme in the blood",
        "If the patient has diabetes",
        "Percentage of blood leaving the heart at each contraction",
        "Woman or man",
        "Platelets in the blood",
        "Level of creatinine in the blood",
        "Level of sodium in the blood",
        "If the patient smokes",
        "Follow-up period",
        "If the patient died during the follow-up period"
    ],
    "Measurement": [
        "Years", "Boolean", "Boolean", "mcg/L", "Boolean", "Percentage", "Binary",
        "kiloplatelets/mL", "mg/dL", "mEq/L", "Boolean", "Days", "Boolean"
    ],
    "Range": [
        "[40, ..., 95]", "0, 1", "0, 1", "[23, ..., 7861]", "0, 1", "[14, ..., 80]",
        "0, 1",
        "[25.01, ..., 850.00]", "[0.50, ..., 9.40]", "[114, ..., 148]", "0, 1",
        "[4, ..., 285]", "0, 1"
    ]
})

data_dict.style.hide()
```

Table 3

Feature	Explanation	Measurement	Range
Age	Age of the patient	Years	[40, ..., 9
Anaemia	Decrease of red blood cells or hemoglobin	Boolean	0, 1
High blood pressure	If a patient has hypertension	Boolean	0, 1
Creatinine phosphokinase (CPK)	Level of the CPK enzyme in the blood	mcg/L	[23, ..., 7
Diabetes	If the patient has diabetes	Boolean	0, 1
Ejection fraction	Percentage of blood leaving the heart at each contraction	Percentage	[14, ..., 8
Sex	Woman or man	Binary	0, 1
Platelets	Platelets in the blood	kiloplatelets/mL	[25.01, ..
Serum creatinine	Level of creatinine in the blood	mg/dL	[0.50, ...,
Serum sodium	Level of sodium in the blood	mEq/L	[114, ...,
Smoking	If the patient smokes	Boolean	0, 1
Time	Follow-up period	Days	[4, ..., 28
Death event	If the patient died during the follow-up period	Boolean	0, 1